

Why design at all?

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Research Design for Social Sciences
MA Computational Social Science, UC3M
Fall 2026

Introduction

- What is this course about?
- Expectations?
- **What is research? And why do we need to design it?**

What this course is, and is not

We do

what variation a design exploits
what has to be true for it to work
how to tell whether an answer is any good

We don't

estimation, models, statistics
code, software, data wrangling
→ that is your methods sequence:
statistics, causal inference,
data science, R, data mining

No statistics in this course at all. Not one equation.

Roadmap

Why design at all?

Three ways to answer a question

Evidence, claims, and inference

Units, observations, variation

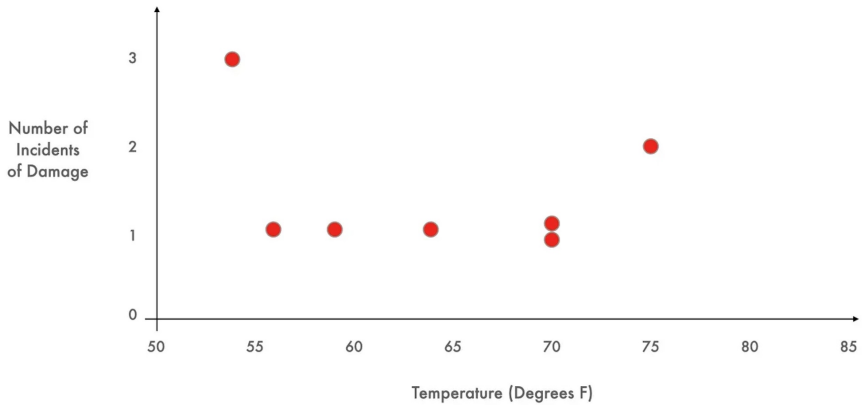
From data to a question

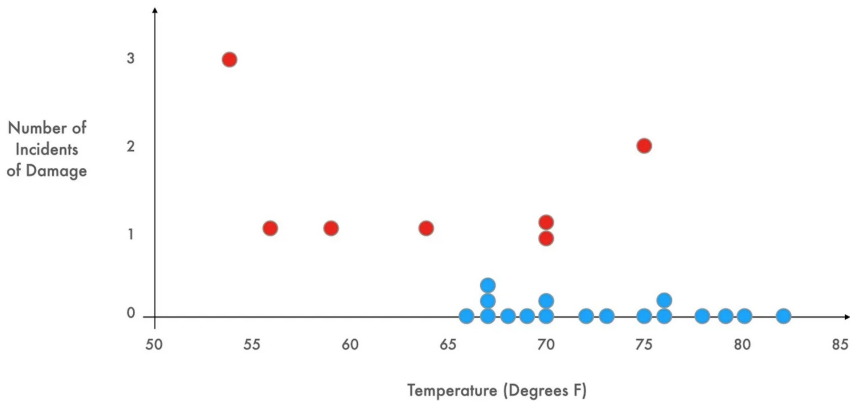
Exercise: audit a claim

Course logistics

A real example

- Racing team deciding whether to race or not in the last session
- Car engine has blown out in 7 out of 24 past races
 - Should we risk an explosion and go bankrupt or race?
- A mechanic has a last-minute gut feeling that it might have to do with temperature (forecast for race: very cold day)
- Someone says: “show me the data of the past failures!”





Space Shuttle *Challenger* disaster

🌐 45 languages ▾

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From Wikipedia, the free encyclopedia

Coordinates:  28°38′24″N 80°16′48″W﻿ / ﻿ 

On January 28, 1986, the [Space Shuttle *Challenger*](#) broke apart 73 seconds into its flight, killing all seven crew members aboard. The spacecraft disintegrated 46,000 feet (14 km) above the Atlantic Ocean, off the coast of [Cape Canaveral](#), Florida, at 11:39 a.m. [EST](#) (16:39 [UTC](#)). It was the first fatal accident involving an [American spacecraft](#) while in flight.

The mission, designated [STS-51-L](#), was the tenth flight for the [orbiter](#) and the twenty-fifth flight of the Space Shuttle fleet. The crew was scheduled to deploy a communications satellite and study [Halley's Comet](#) while they were in orbit, in addition to taking school teacher [Christa McAuliffe](#) into space. The latter resulted in a higher than usual media interest and coverage of the mission; the launch and subsequent disaster were seen live in many schools across the United States.

The cause of the disaster was the failure of the primary and secondary redundant [O-ring](#) seals in a joint in the shuttle's right [solid rocket booster](#) (SRB). The record-low temperatures on the morning of the launch had stiffened the rubber O-rings, reducing their ability to seal the joints. Shortly after liftoff, the seals were breached, and hot pressurized gas from within the SRB leaked through the joint and burned through the aft attachment strut connecting it to the [external propellant tank](#) (ET), then into the tank itself. The [collapse](#) of the ET's internal structures and the rotation of the SRB that followed threw the shuttle

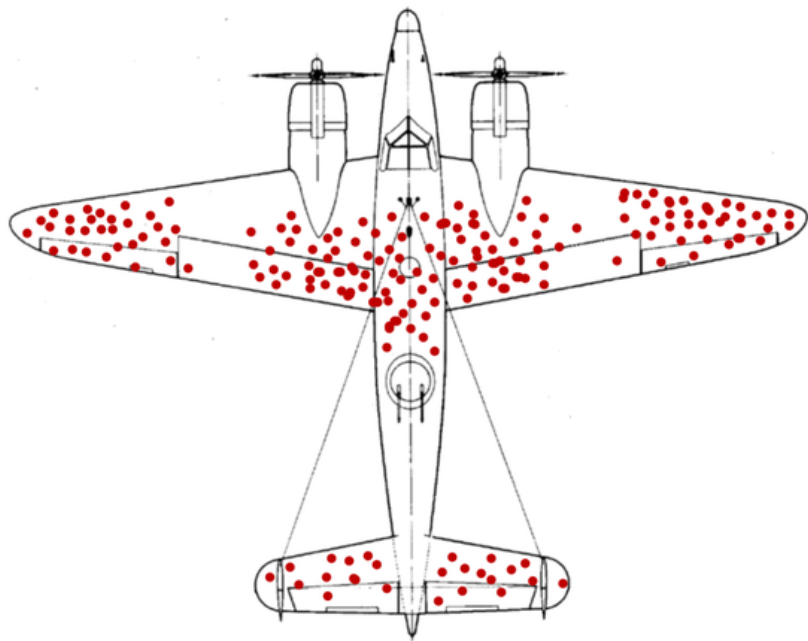
Space Shuttle *Challenger* disaster



Challenger's [solid rocket boosters](#) fly uncontrollably after the breakup of the [external tank](#) separated them from the shuttle stack. The remains of the orbiter and tank leave thin white contrails as they fall toward the Atlantic Ocean.

Challenger example

- Explanatory research
- What is the role of design here?
- Which observations to use
- What *variation* we need to answer the question?



Not only about data or empirical evidence

- The “non-data” part is also important:
thinking about an answerable question, a coherent argument, and how to connect it to data
- Also problematic because we all probably think we already know how it's done, but e.g. many final theses fail on this

Types of research

- *Theoretical* and *empirical* research
- *Qualitative* and *quantitative* empirical research
 - Descriptive vs explanatory

Roadmap

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From data to a question

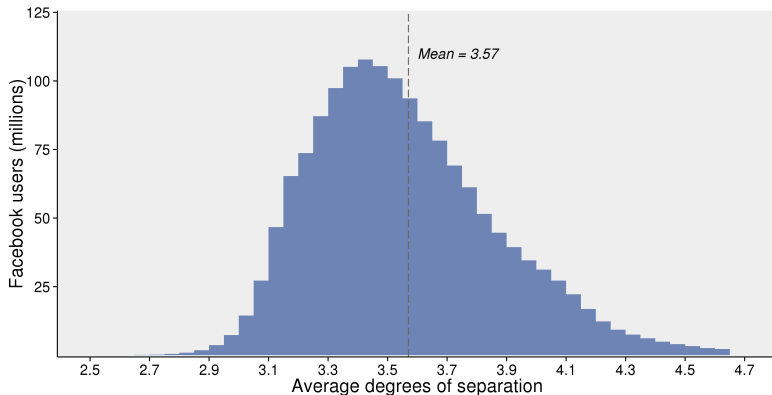
Exercise: audit a claim

Course logistics

Three things you can do with data

- **Describe:** what is the case? How much, where, for whom?
- **Explain:** does X cause Y , and through what?
- **Predict:** given what I observe, what happens next, or elsewhere?

Describe



1.59bn Facebook users; average degrees of separation = 3.57.

No causal claim, no prediction, and still full of design decisions.

SOCIAL MEDIA

How do social media feed algorithms affect attitudes and behavior in an election campaign?

Andrew M. Guess^{1*}, Neil Malhotra², Jennifer Pan³, Pablo Barberá⁴, Hunt Allcott⁵, Taylor Brown⁴, Adriana Crespo-Tenorio⁴, Drew Dimmery^{4,6}, Deen Freelon⁷, Matthew Gentzkow⁸, Sandra González-Bailón⁹, Edward Kennedy¹⁰, Young Mie Kim¹¹, David Lazer¹², Devra Moehler⁴, Brendan Nyhan¹³, Carlos Velasco Rivera⁴, Jaime Settle¹⁴, Daniel Robert Thomas⁴, Emily Thorson¹⁵, Rebekah Tromble¹⁶, Arjun Wilkins⁴, Magdalena Wojcieszak^{17,18}, Beixian Xiong⁴, Chad Kiewiet de Jonge⁴, Annie Franco⁴, Winter Mason⁴, Natalie Jomini Stroud¹⁹, Joshua A. Tucker²⁰

We investigated the effects of Facebook's and Instagram's feed algorithms during the 2020 US election. We assigned a sample of consenting users to reverse-chronologically-ordered feeds instead of the default algorithms. Moving users out of algorithmic feeds substantially decreased the time they spent on the platforms and their activity. The chronological feed also affected exposure to content: The amount of political and untrustworthy content they saw increased on both platforms, the amount of content classified as uncivil or containing slur words they saw decreased on Facebook, and the amount of content from moderate friends and sources with ideologically mixed audiences they saw increased on Facebook. Despite these substantial changes in users' on-platform experience, the chronological feed did not significantly alter levels of issue polarization, affective polarization, political knowledge, or other key attitudes during the 3-month study period.

Feeds randomly switched from algorithmic to chronological, 3 months, $N \approx 23,000$.

Randomisation buys the comparison. It does not buy generality.

ECONOMICS

Predicting poverty and wealth from mobile phone metadata

Joshua Blumenstock,^{1*} Gabriel Cadamuro,² Robert On³

Accurate and timely estimates of population characteristics are a critical input to social and economic research and policy. In industrialized economies, novel sources of data are enabling new approaches to demographic profiling, but in developing countries, fewer sources of big data exist. We show that an individual's past history of mobile phone use can be used to infer his or her socioeconomic status. Furthermore, we demonstrate that the predicted attributes of millions of individuals can, in turn, accurately reconstruct the distribution of wealth of an entire nation or to infer the asset distribution of microregions composed of just a few households. In resource-constrained environments where censuses and household surveys are rare, this approach creates an option for gathering localized and timely information at a fraction of the cost of traditional methods.

Phone metadata → wealth estimates for a whole country.

Identifies nothing causal, and does not need to. It has to work *out of sample*.

Different goals, different standards

Goal	Succeeds when	Typical failure
Describe	coverage, comparability, a defensible denominator	measuring what got recorded, not what happened
Explain	the comparison isolates the effect	something else moved at the same time
Predict	it works out of sample	it worked, until the world changed

- Most bad designs are a **mismatch**: a descriptive question with a causal answer bolted on, or a causal question answered with a predictive design

The five questions of this course

1. **Why design at all?** (today)
2. **What is the question, and where would the data come from?**
3. **Do the numbers mean what you think they mean?**
4. **What comparisons can we use to establish causal claims?**
5. **What variation can you exploit, and how far does the answer travel?**

By the workshop, you should be able to answer all five *about your own project*.

Roadmap

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Empirical research

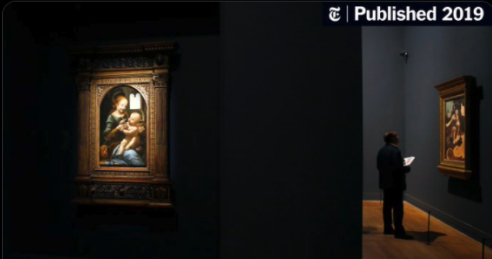
- Goal: answer a question using empirical evidence
 - Usual problems: unanswerable questions, wrong data to question, concepts do not correspond to measurements...
- How? exploiting **empirical variation**
- Research design is essentially knowing where to look at, how to measure it, how to analyze it, how to interpret it, etc –
it's about **inference**
using what you *did* observe to say something about what you did *not*
- Two things we should learn:
how to answer questions & how to evaluate answers

Empirical evidence and claims

 **NYT Health** 
@NYTHealth

Want to live longer? Try going to the opera.
Researchers in Britain have found that people who reported going to a museum or concert even once a year lived longer than those who didn't.

 | Published 2019



Another Benefit to Going to Museums? You May Live Longer (Published 2019)
Researchers in Britain found that people who go to museums, the theater and the opera were less likely to die in the study period than those who didn't.
[nytimes.com](https://www.nytimes.com)

3:19 PM · Dec 22, 2019 · SocialFlow

Empirical evidence and claims

- Let's reverse-engineer this:
 - What did they probably do?
- How would you answer the question better?
- Causal inference and experimental methods
 - Observation vs manipulation
- Gold standard?



REPORT



Analytic Thinking Promotes Religious Disbelief

WILL M. GERVAIS AND ARA NORENZAYAN [Authors Info & Affiliations](#)

SCIENCE • 27 Apr 2012 • Vol 336, Issue 6080 • pp. 493-496 • DOI: 10.1126/science.1215647



1,735



CHECK ACCESS

Overthinking Religion?

Many theories of human cognition make a distinction between System I, which tends to be rapid and to rely on heuristics or rules of thumb, and System II, which tends to be more deliberative and analytic. This dual-process framework, within which both processes may operate simultaneously and competitively, has been used to explain a variety of situational influences upon decision-making. **Gervais and Norenzayan** (p. 493) studied the application of a dual-process framework to religious disbelief and found that triggering analytic thinking processes through a variety of experimental manipulations resulted in a tendency for subjects to report lower levels of religious belief.

Overthinking Religion? |

Abstract

Full Text Material

Supplemental Notes

eLetters (0)



Opinions about claims?

Table 2. Summary of experimental methods and findings (studies 2 to 5). *d* reflects effect size estimates (Cohen's *d*).

Study Belief measure (possible range)	Condition: sample stimuli	<i>N</i>	<i>M</i>	<i>SD</i>	<i>t</i>	<i>P</i>	<i>d</i>
2: Art	Control: <i>Discobolus</i>	31	61.55	35.68	2.24	0.03	0.59
Belief in God (0–100)	Analytic: <i>The Thinker</i>	26	41.42	31.47			
3: Implicit	Control: hammer, shoes, jump, retrace, brown	43	12.65	5.29	2.11	0.04	0.44
Supernatural agents (3–21)	Analytic: think, reason, analyze, ponder, rational	50	10.12	6.13			
4: Implicit	Control: hammer, shoes, jump, retrace, brown	75	40.16	16.73	2.20	0.03	0.36
Intrinsic religiosity (10–70)	Analytic: think, reason, analyze, ponder, rational	70	34.39	14.77			
5: Disfluency	Control: sample font	88	12.16	5.99	2.06	0.04	0.31
Supernatural agents (3–21)	Analytic: <i>sample font</i>	91	10.40	5.44			

‘Your view of human nature will change
profoundly as you read this brilliant book.’

DANIEL KAHNEMAN
*Author of *Thinking, Fast and Slow**

THE MARSHMALLOW TEST

**Understanding self-control and
how to master it**



WALTER MISCHEL

Watts and his colleagues were skeptical of that finding. The original results were based on studies that included fewer than 90 children—all enrolled in a preschool on Stanford's campus. In restaging the experiment, Watts and his colleagues thus adjusted the experimental design in important ways: The researchers used a sample that was much larger—more than 900 children—and also more representative of the general population in terms of race, ethnicity, and parents' education. The researchers also, when analyzing their test's results, controlled for certain factors—such as the income of a child's household—that might explain children's ability to delay gratification and their long-term success.

Ultimately, the new study finds limited support for the idea that being able to delay gratification leads to better outcomes. Instead, it suggests that the capacity to hold out for a second marshmallow is shaped in large part by a child's social and economic background—and, in turn, that that background, not the ability to delay gratification, is what's behind kids' long-term success.

Other types of inference

- We are used to cases where we compare many observations with different values on key variables
- What if we don't have enough observations to answer a question?
- Any idea about examples?

Europe

Violent storm kills six on Corsica as island raises new alert

By **Marc Angrand** and **Benoit Van Overstraeten**

August 18, 2022 8:34 PM GMT+2 · Updated a year ago



Aa



PARIS, Aug 18 (Reuters) - A violent and unexpected storm battered the French Mediterranean island of Corsica on Thursday, killing at least six people including a teenage girl, and meteorologists predicted more bad weather to come.

Hail, heavy rain and winds peaking at 224 km per hour (140 mph) swept the island early in the day. Two of the victims were killed when trees fell in campsites.

"Storms formed at sea will affect large parts of the western Corsica coast throughout the night from Thursday to Friday," Meteo France forecaster said.

Opinions? How do you think it's done?

Article Type: **Research Article**

 Open access

[View license](#)

Anthropogenic Warming Had a Crucial Role in Triggering the Historic and Destructive Mediterranean Derecho in Summer 2022

Juan Jesús González-Alemán, Damián Insua-Costa, Eric Bazile, Sergi González-Herrero, Mario Marcello Miglietta, Pieter Groenemeijer, and Markus G. Donat

Online Publication: **31 Aug 2023**

Print Publication: **01 Aug 2023**

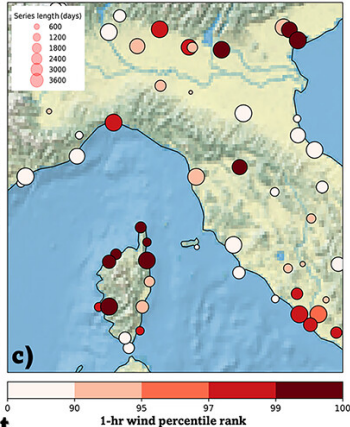
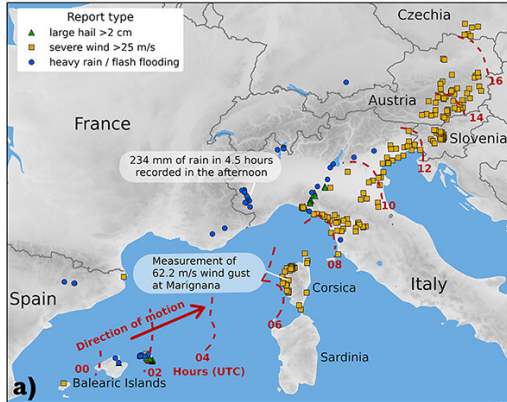
DOI: <https://doi.org/10.1175/BAMS-D-23-0119.1>

Page(s): **E1526–E1532**

[Article History](#)

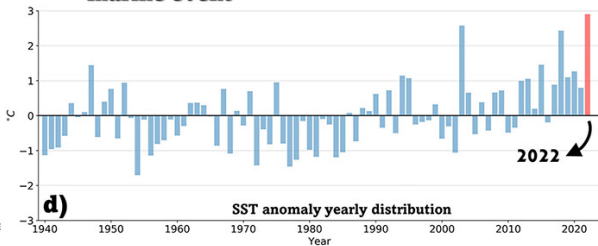
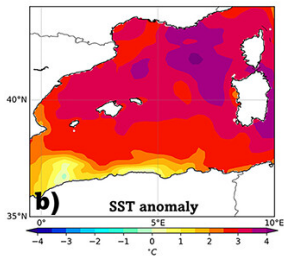
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atmospheric event

marine event



A few questions on variation

- Why do we talk so much about *variation*?
- Critique of having “few observations”?
- Why do we use statistics at all?
- What about experiments?
- Where's the variation in attribution science?

Halley's Eclipse: a coup for Newtonian prediction and the selling of science

300 years ago, on the 3rd of May 1715, a rare solar eclipse occurred over England. It was an opportunity too good to miss for those promoting new astronomical theories - and their own careers.



A few questions on variation

- Why do we talk so much about *variation*?
- Critique of having “few observations”?
- Why do we use statistics at all?
- What about experiments?
- Where's the variation in attribution science?

Description is not easy either

- Are levels of political violence increasing?
- Is the world becoming less democratic?

Roadmap

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Exercise: audit a claim

Course logistics

Key ingredients

A dataset, and the four words we will use every week:

unit of
analysis:
country-year

country	year	GDP p/c
Spain	2020	27,100
Spain	2021	30,100
France	2020	39,000
France	2021	43,700

one **observation**

a **variable**

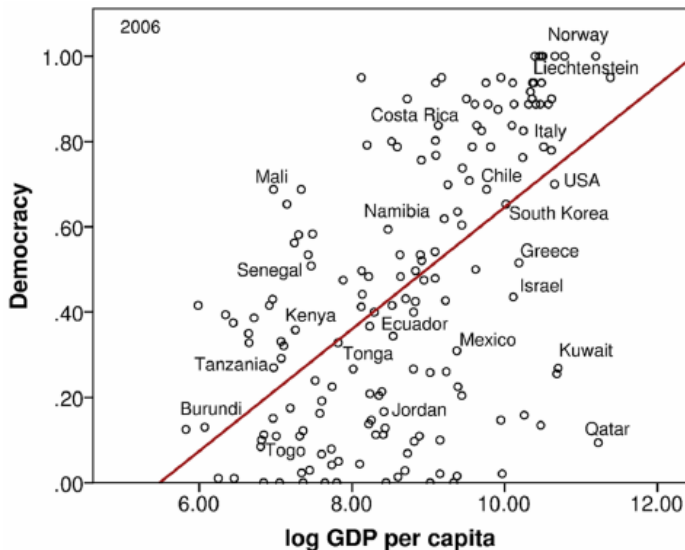
Variation = the differences *down* a column that a comparison can use.

No variation, nothing to learn.

On the unit of analysis

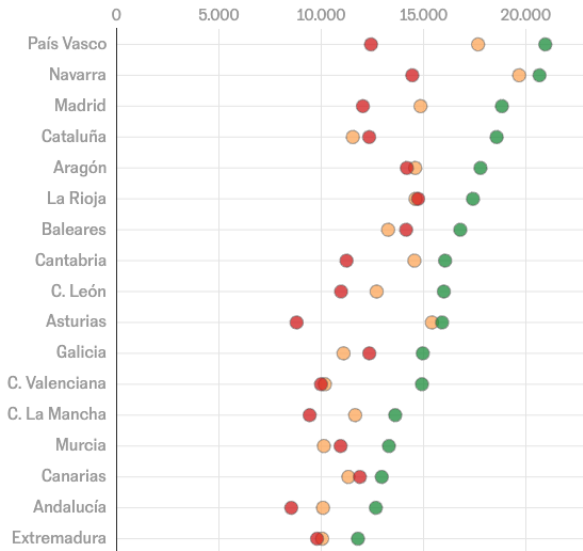
- Why is this important?
- Able to quickly identify it?

What's the unit of analyses in the data behind this graph?



Renta mediana de por unidad de convivencia de cada tipo de familia en cada comunidad autónoma

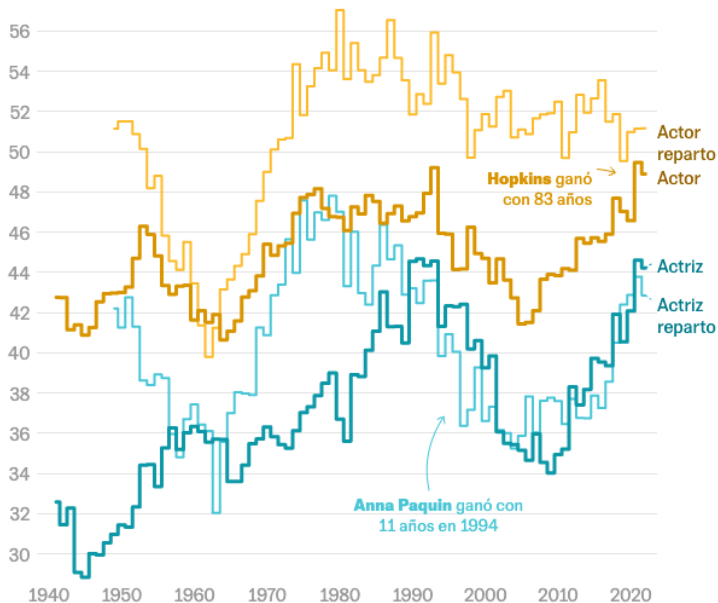
● Todos ● Familia numerosa ● Un adulto con 1 o 2 niños dep



Los datos de Extremadura, Aragón, Canarias, Cantabria, La Rioja y Asturias cuentan solo con entre 50 y 100 entrevistas

ECV 2019, 2020 y 2021 (INE) / EL PAÍS

Edad de los actores premiados (media móvil de 12 años)



On the unit of analysis

- Most empirical designs can be adapted to different units of analyses
- The unit you use is related to the question and theory
- Example: children's educational attainment and contextual factors (peers, school, family, etc)
 - How many different units could you use, and how do they map onto different questions?

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Where does your data come from?

Designed

surveys, experiments,
a coding scheme you wrote

→ answers a question **you** had

Found

digital traces, admin records,
scraped text, API dumps

→ answers a question **someone else**
had.

Or no question at all

A lot of what you will work with in this MA is **found data**

(Technically, when you use existing surveys or datasets that's also found data, but it is not the same as in typical CSS data, because it was collected with a research design in mind, and that's usually documented)

The problem of starting with the data first

Two examples, both started the way most of yours will start: **with data already in hand** and **no question yet** (which shouldn't be the case, but...)

- A hotel's booking records
- Spotify's collaboration network

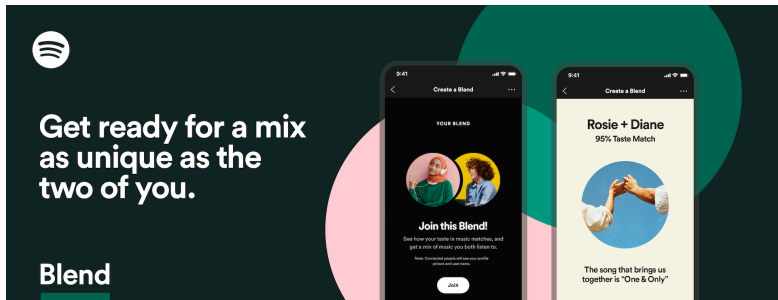
Example 1: hotel bookings

- Data arrives from a hotel: every booking, and whether the guest actually showed up
- Unit of analysis: **the booking**
- **Motivation:** we can use data to increase business
- **Research question:** _____?

Example 1: what can these data answer?

- Which **goal** could this be: describe, explain, or predict?
- What would each version actually look like?
- Is “the booking” the right **unit**? What else could it be?
- Who is **missing** from this data?
- And: is the question you can answer the one the hotel wanted?

Example 2: music collaborations



- Spotify API data
- Build a turned a **network** of artist collaborations
- Variables: time, genre

Example 2: what can we ask here?

- What is the unit of analysis? (before and after?)
- What questions can we answer here? General design approach?
- Problems of data origin (e.g are genres becoming more connected over time or catalogue issues? who decides how to tag artists?)

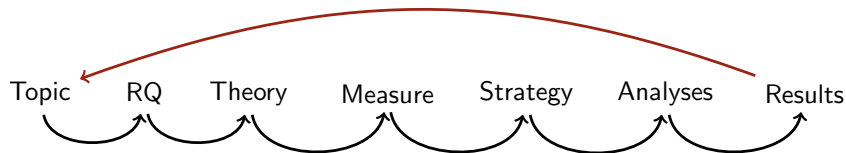
The Parable of Google Flu: Traps in Big Data Analysis

What both have in common

- Both are **found** data
- Both entered the process **in the middle** (at the data, not at the question)
- Neither dataset came with a **goal**, a **unit**, or a **question**

That is what the rest of this course supplies.

The research process



The red arrow is the honest part: nobody does this once, in order.

From topic to question, and from question to theory

- > From **topic** to the **research question**
 - Motivation is not a question
 - Good ones are answerable and relevant
- > Examples, good and bad:
 - What is the best Netflix show?
 - What can we do to help poor countries develop?
 - What are the shopping patterns of the Spanish population?
 - Do individuals from minorities support the use of violence?
- > From **RQ** to **theory**: arguments and mechanisms, testable and credible

And the rest of the process

- > From **theory** to **measurement**: concepts, operationalization, unit of analysis (week 3)
- > From **measurement** to **strategy**: what variation will you look at? (weeks 4–5)
- > From **strategy** to **analyses**: *not covered in this course*
 - **Question**: what is the role of statistics in research design?
- > From **analyses** to **results**: interpretation, relevance, **external validity**

Simplifying

1. Formulate one (or more) RQ from a topic/problem
2. Develop a theoretical argument related to that RQ
3. Building on theory, think about what and how to measure
4. What variation are you going to look at?
5. What can you really learn from the results?

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Exercise: audit a claim

Course logistics

Exercise: audit a claim

- A. “Remote work has made young people lonelier.”
- B. “Four in ten Spanish adults under 30 still live with their parents.”
- C. “Our model flags students at risk of dropping out with 87% accuracy.”

(invented headlines)

In pairs, **your assigned claim only**. One written sentence each:

1. What is the **goal**: describe, explain, or predict?
2. What would the **data** have to look like for this to be knowable?
3. What is the **first thing** you would doubt?

10 min in pairs, then 15 min together.

What we just did

- Named the **goal** before arguing about the answer
- Asked what data would have to exist
- Found the weak point in the **design**, not in the finding

This is the whole course, in miniature. You will do it to a published paper every week, and to your own project in the workshop.

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Course logistics

- Tuesdays 18:00–21:00, except the workshop: **Mon Oct 26, 15:00–21:00**
- Six sessions between today and October 26th
- My email: `francisco.villamil@uc3m.es`
- Office hours by appointment
- https://franvillamil.github.io/res_design/

Calendar

- **Sep 15:** Why design at all?
- **Sep 22:** Questions, and where the data comes from
- **Sep 29:** Do the numbers mean what you think?
- **Oct 6:** What comparisons can we use to establish causal claims?
- **Oct 13:** What variation can you exploit, and how far does it travel?
- **Oct 26** (Monday, 15h–21h): **Workshop**

Papers discussed in sessions 2, 3 and 4. Attendance is mandatory.

Evaluation

- Design memos (15%) pair
- **Workshop presentation (35%)** group
- Workshop feedback on your paired group (10%) group
- Final essay (40%) group

No participation grade. No graded paper commentary.

45% is earned on the workshop day.

The design memos

	Due	Task
M1	Mon Sep 21	Audit a claim: the goal, the data it would need, your first doubt
M2	Mon Sep 28	A found data source: who made it, who's missing, what it can answer
M3	Mon Oct 5	One concept, operationalised two ways. Which, and what you lose
M4	Mon Oct 12	A causal claim: the comparison that would license it, and what would bias it
M5	Mon Oct 19	Your group's project , goes to your paired group

- In pairs, half a page. Aula Global, **Monday night** (formally Tue 00.00h)
- M1–M4 are **not** about your essay, but practice on someone else's problem. M5 is your group's, and it's the exception

Papers

1. **Sep 22:** Lazer, Kennedy, King & Vespignani (2014) **The Parable of Google Flu: Traps in Big Data Analysis.** *Science* 343(6176).
 2. **Sep 29:** Müller-Crepon, Hunziker & Cederman (2021) **Roads to Rule, Roads to Rebel.** *JCR* 65(2–3).
 3. **Oct 6:** Guess *et al.* (2023) **How do social media feed algorithms affect attitudes and behavior in an election campaign?** *Science* 381(6656).
- **Nothing to hand in, no mark.** But part of the session is a conversation about a paper you'll have to have read

The workshop

- **Oct 26**, 15h–21h. Groups of 3–4, 12 min + 8 min discussion
- Required slides: the goal; the question; units and variation; the concept and its measure; the comparison; and **the threat I can't solve**
- Each group is **paired with another**: you read their design sketch and give them written feedback, and they do the same for you
- Groups are fixed by **Mon Oct 19**, when Memo 5 is due. Whoever submits Memo 5 together presents together

The final essay

- **In groups** of 3–4 (the same group you present with). Max **5000 words**
- A **pre-analysis plan**: the document a researcher actually writes *before* collecting anything
- **Deadline: Oct 27**, 23.59h, on Aula Global
- It should be **finished** before you present.

The extra day is for changes, not for writing it

Books

- **Matthew Salganik, *Bit by Bit: Social Research in the Digital Age* (Princeton UP, 2018).** Free online. Closest to this course
- **Nick Huntington-Klein, *The Effect* (Chapman and Hall/CRC, 2021).** Free online. Ch. 1–6 for us
- **Dimiter Toshkov, *Research Design in Political Science* (Palgrave, 2016)**
- **Kosuke Imai, *Quantitative Social Science* (Princeton UP, 2017)** methods courses
- **Scott Cunningham, *Causal Inference: The Mixtape* (2021)** causal inference

Before next week

- **Read:** Lazer *et al.*, *The Parable of Google Flu* (three pages, no equations)
 - Nothing to hand in. We discuss it next week
- **M1**, in pairs, on Aula Global before next class:
 - Take a claim, from the news or anywhere
 - Name the goal. Say what data it would need. Name the first thing you'd doubt
 - Half a page

Where we started

“The answer was in the observations
they didn't look at.”

1. Why design at all?

done

2–5: the question, the measure, the comparison, how far it travels

Next week: **where does the data come from?**

Questions?

Roadmap

Appendix

Evaluating the replicability of social science experiments in *Nature* and *Science* between 2010 and 2015

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Being able to replicate scientific findings is crucial for scientific progress¹⁻¹⁵. We replicate 21 systematically selected experimental studies in the social sciences published in *Nature* and *Science* between 2010 and 2015¹⁶⁻³⁶. The replications follow analysis plans reviewed by the original authors and pre-registered prior to the replications. The replications are high powered, with sample sizes on average about five times higher than in the original studies. We find a significant effect in the same direction as the original study for 13 (62%) studies, and the effect size of the replications is on average about 50% of the original effect size. Replicability varies between 12 (57%) and 14 (67%) studies for complementary replicability indicators. Consistent with these results, the estimated true-

a significant effect in the same direction as the original studies for 61% of replications¹³. Both the RPP and the EERP had high statistical power to detect the effect sizes observed in the original studies. However, the effect sizes of published studies may be inflated even for true-positive findings owing to publication or reporting biases⁴⁰⁻⁴². As a consequence, if replications were well powered to detect effect sizes smaller than those observed in the original studies, replication rates might be higher than those estimated in the RPP and the EERP.

We provide evidence about the replicability of experimental studies in the social sciences published in the two most prestigious general science journals, *Nature* and *Science* (the Social Sciences Replication Project (SSRP)). Articles published in these